

DAY 2 PRACTICE TASKS ON PUBLIC IP vs PRIVATE IP ADDRESSING

✓ TASK 1 — PRIVATE IP ADDRESS IDENTIFICATION

📌 Task Name

Private IP Address Identification using IPConfig

🎯 Objective

To identify the system's private IPv4 address, subnet mask, and default gateway.

✂ Steps Performed

Step 1: Opened Command Prompt.

Step 2 Executed command: ipconfig

Step 3

Observed:

- IPv4 Address
- Subnet Mask
- Default Gateway

Output/Screenshot

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::cc07:b083:fd63:5f29%6
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:97c2:aa1e:ebf1:ff46:da43:a723
    Temporary IPv6 Address. . . . . : 2401:4900:97c2:aa1e:580b:4c41:e196:417c
    Link-Local IPv6 Address . . . . . : fe80::c1bc:c3f0:2af5:fef9%7
    IPv4 Address. . . . . : 10.221.64.26
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::90b0:c0ff:fe1e:8324%7
                                10.221.64.104

C:\Users\Akshaya>
```

📊 Result

Successfully identified the local/private IPv4 address and network configuration details.

Learning Outcome

Learned how internal devices are identified inside a private network.

Important Notes for IAM Analysts

IAM systems use private IP information to:

- Identify trusted internal devices
- Monitor internal network access
- Apply internal access policies

TASK 2 — PUBLIC IP ADDRESS IDENTIFICATION

Task Name

Public IP Address Identification

Objective

To identify the internet-visible public IP address assigned by ISP.

Steps Performed

Step 1

Opened web browser.

Step 2

Searched:

What is my IP

Step 3

Observed:

- Public IP
- ISP Name
- Region

Output/Screenshot

The screenshot shows the homepage of 'whatismyipaddress.com'. The main content area displays the user's IP information:

- My IP Address is:**
 - IPv6: `2401:4900:97c2:aa1e:81d:73e6:3dec:8be0`
 - IPv4: `106.192.36.233`
- My IP Information:**
 - ISP: Bharti Airtel Ltd.
 - City: Yanam
 - Region: Andhra Pradesh
 - Country: India
- Your location may be exposed!** (with a red button: **HIDE MY IP ADDRESS NOW**)
- Map:** A map showing the location near Visakhapatnam, India, with coordinates `2401:4900:97c2:aa1e:81d:73e6:3dec:8be0`.

There are also Canva advertisements and a search bar at the top.

Result

Successfully identified public internet-facing IP address.

Learning Outcome

Learned difference between internet-visible and internal IP communication.

Important Notes for IAM Analysts

Public IPs are monitored for:

- Geo-location analysis
- Risk-based authentication
- Suspicious login detection

TASK 3 — PRIVATE IP RANGE ANALYSIS

Task Name

Private IP Range Verification

Objective

To verify which RFC1918 private IP range the system belongs to.

Steps Performed

Step 1

Observed IPv4 address from ipconfig.

Step 2

Compared with private ranges:

10.x.x.x

172.16.x.x – 172.31.x.x

192.168.x.x

Step 3

Documented IP class.

Output/Screenshot

```
Link-local IPv6 Address . . . . . : fe80::e1bc:ec3b:2a7b:7c19%7  
IPv4 Address. . . . . : 10.221.64.26
```



Result

Successfully identified private IP category and address range.



Learning Outcome

Learned classification of enterprise private networks.



Important Notes for IAM Analysts

Organizations use private IP ranges to:

- Segment departments
- Restrict internal access
- Secure enterprise communication



TASK 4 — INTERNET CONNECTIVITY TEST



Task Name

Internet Connectivity Testing using Ping



Objective

To test communication between local machine and internet server.

✂ Steps Performed

Step 1

Opened Command Prompt.

Step 2

Executed:

ping google.com

Step 3

Observed:

- Packet replies
- Response time
- Packet loss

Output/Screenshot

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ping google.com

Pinging google.com [2404:6800:4007:833::200e] with 32 bytes of data:
Reply from 2404:6800:4007:833::200e: time=58ms
Reply from 2404:6800:4007:833::200e: time=115ms
Reply from 2404:6800:4007:833::200e: time=92ms
Reply from 2404:6800:4007:833::200e: time=51ms

Ping statistics for 2404:6800:4007:833::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 51ms, Maximum = 115ms, Average = 79ms

C:\Users\Akshaya>
```



Result

Successfully verified internet connectivity.



Learning Outcome

Learned how systems verify external communication.



Important Notes for IAM Analysts

Ping testing helps IAM/security teams:

- Verify VPN connectivity
- Troubleshoot access issues
- Detect communication failures



TASK 5 — PUBLIC IP CONNECTIVITY TEST



Task Name

Direct Public IP Connectivity Test



Objective

To verify internet connectivity using direct public IP communication.

✂ Steps Performed

Step 1

Opened Command Prompt.

Step 2

Executed:

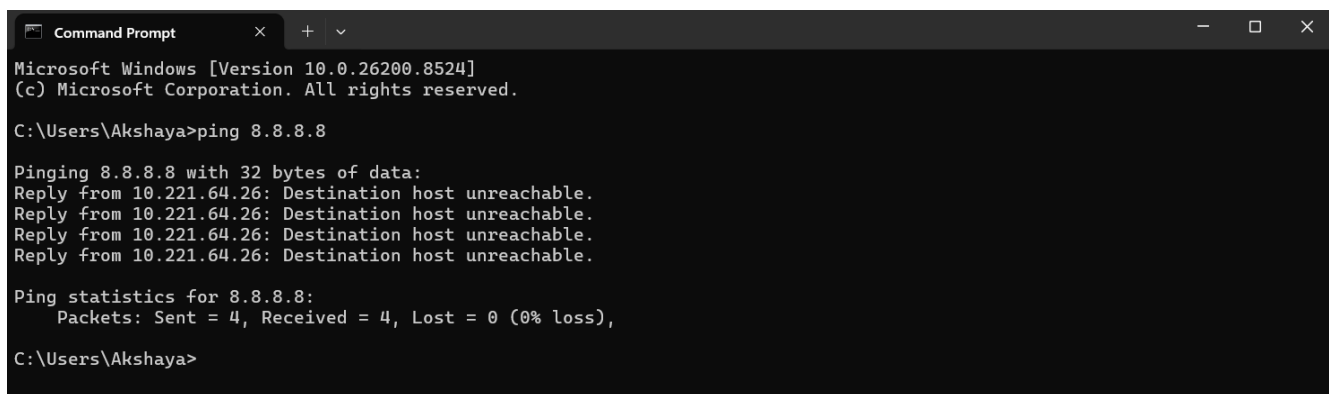
ping 8.8.8.8

Step 3

Observed:

- Packet transmission
- Internet reachability

Output/Screenshot



```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 10.221.64.26: Destination host unreachable.
Reply from 10.221.64.26: Destination host unreachable.
Reply from 10.221.64.26: Destination host unreachable.
Reply from 10.221.64.26: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Akshaya>
```

Result

Successfully verified direct communication with external public server.

Learning Outcome

Learned difference between DNS-related and direct IP communication.

Important Notes for IAM Analysts

Helps identify:

- DNS failures
 - Routing issues
 - Firewall restrictions
-

✓ TASK 6 — DNS RESOLUTION ANALYSIS

📌 Task Name

DNS Lookup Analysis using NSLookup

🎯 Objective

To analyze how DNS converts domain names into IP addresses.

✂ Steps Performed

Step 1

Opened Command Prompt.

Step 2

Executed:

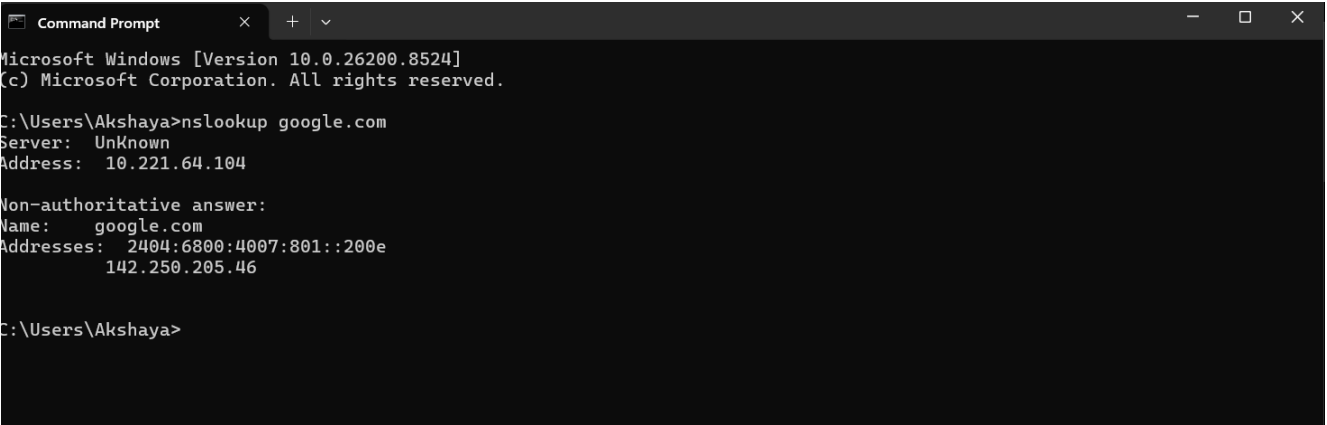
```
nslookup google.com
```

Step 3

Observed:

- DNS server
- Resolved IP addresses

Output/Screenshot



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>nslookup google.com
Server: UnKnown
Address: 10.221.64.104

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4007:801::200e
          142.250.205.46

C:\Users\Akshaya>
```

📊 Result

Successfully resolved domain into IP addresses.

🧠 Learning Outcome

Learned DNS communication and internet name resolution.

Important Notes for IAM Analysts

IAM systems rely on secure DNS communication for:

- Authentication services
- Cloud access
- Identity federation

TASK 7 — NETWORK ROUTE ANALYSIS

Task Name

Network Route Tracing using TraceRoute

Objective

To analyze how packets travel across networks.

Steps Performed

Step 1

Opened Command Prompt.

Step 2

Executed:

tracert google.com

Step 3

Observed:

- Routers
- Hops
- Packet route

Output/Screenshot

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>tracert google.com

Tracing route to google.com [2404:6800:4007:83f::200e]
over a maximum of 30 hops:

  1    3 ms    2 ms    2 ms  2401:4900:9373:5beb::6e
  2   117 ms   98 ms   99 ms  2401:4900:1:a809::2
  3   103 ms  100 ms   97 ms  2401:4900:1:a882::3
  4   106 ms  100 ms   98 ms  2401:4900:1:a883::1
  5    *      *      *      Request timed out.
  6    53 ms  151 ms   21 ms  2404:a800:3a00:2::79
  7    *      *      *      Request timed out.
  8  1004 ms   68 ms   69 ms  2001:4860:1:1::674
  9    44 ms   36 ms   54 ms  2001:4860:0:1::a03d
 10    44 ms   44 ms   39 ms  2001:4860:0:1::2d47
 11    53 ms   36 ms   47 ms  pnmaaa-at-in-x0e.1e100.net [2404:6800:4007:83f::200e]

Trace complete.

C:\Windows\System32>
```


Result

Successfully traced internet communication path.

Learning Outcome

Learned packet routing process and network traversal.

Important Notes for IAM Analysts

Used for:

- Troubleshooting authentication delays
- Network diagnostics
- Security investigations

TASK 8 — ACTIVE NETWORK CONNECTION ANALYSIS

Task Name

Network Session Analysis using NetStat

Objective

To view active TCP/IP network sessions.

Steps Performed

Step 1

Opened Command Prompt.

Step 2

Executed:

netstat

Step 3

Observed:

- Active connections
- Ports
- Listening services

Output/Screenshot

```
C:\Users\Akshaya>netstat
Active Connections

```

Proto	Local Address	Foreign Address	State
TCP	10.221.64.26:59758	13.204.106.34:https	ESTABLISHED
TCP	127.0.0.1:49671	DESKTOP-V8SEPE5:49672	ESTABLISHED
TCP	127.0.0.1:49672	DESKTOP-V8SEPE5:49671	ESTABLISHED
TCP	127.0.0.1:59756	DESKTOP-V8SEPE5:59757	ESTABLISHED
TCP	127.0.0.1:59757	DESKTOP-V8SEPE5:59756	ESTABLISHED

```
C:\Users\Akshaya>
```

Result

Successfully identified active network communication sessions.

Learning Outcome

Learned how systems maintain network sessions.

Important Notes for IAM Analysts

Helps monitor:

- Unauthorized connections
- Suspicious ports
- Malicious communication

TASK 9 — PUBLIC IP GEOLOCATION ANALYSIS

Task Name

Public IP Geo-location Investigation

Objective

To identify geographical information associated with public IP.

Steps Performed

Step 1

Opened:

[IP Lookup Tool](#)

Step 2

Entered public IP.

Step 3

Observed:

- Country

- Region
- ISP

Output/Screenshot

106.192.36.233		IP Lookup
IP Address	106.192.36.233	
City	Hyderabad	Update
Region	Telangana	
Country	IN India	
Postal Code	500055	
European Union	false	
Latitude / Longitude	17.3843 , 78.4583	
Time Zone	Asia/Kolkata (+0530)	
Calling Code	+91	
Currency	INR	
Languages	en-IN,hi,bn,te,mr,ta,ur,gu,kn,ml,or,pa,as,bh...	
ASN	AS45609	
Org	Bharti Airtel Ltd. AS for GPRS Service	

Locate my IP on Google Maps

Result

Successfully identified geographical information linked to public IP.

Learning Outcome

Learned how organizations detect suspicious login locations.

Important Notes for IAM Analysts

Geo-location analysis helps:

- Detect impossible travel
- Trigger MFA
- Block risky logins

TASK 10 — NAT COMMUNICATION FLOW ANALYSIS

Task Name

NAT Flow Architecture Analysis

Objective

To understand how routers translate private IP into public IP.

✂ Steps Performed

Step 1

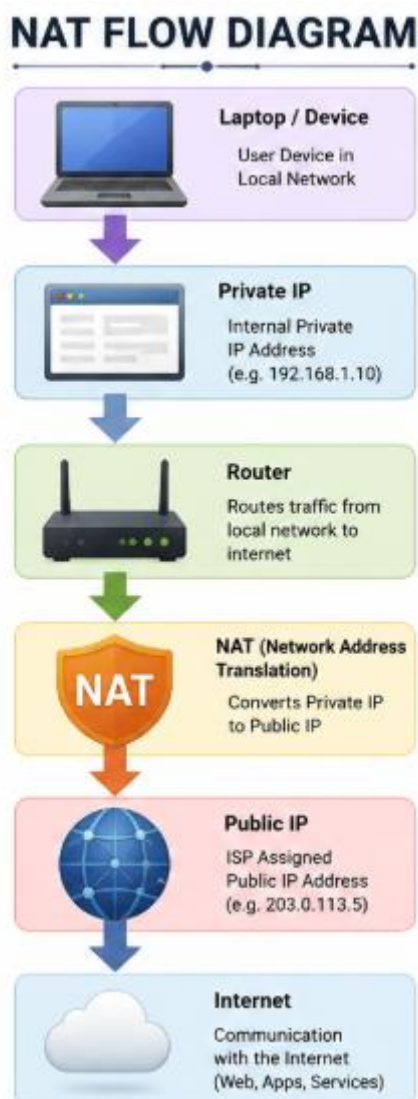
Created NAT communication diagram.

Step 2

Mapped:

Private IP → Router → NAT → Public IP

Output/Screenshot



📊 Result

Successfully analyzed NAT translation process.

🧠 Learning Outcome

Learned how internet communication works through NAT.

Important Notes for IAM Analysts

NAT helps organizations:

- Hide internal networks
- Improve security
- Control internet exposure

TASK 11 — IAM ACCESS CONTROL FLOW ANALYSIS

Task Name

IAM IP-Based Access Control Analysis

Objective

To understand how IAM systems use IP intelligence during authentication.

Steps Performed

Step 1

Created IAM authentication flow diagram.

Step 2

Included:

IP Check

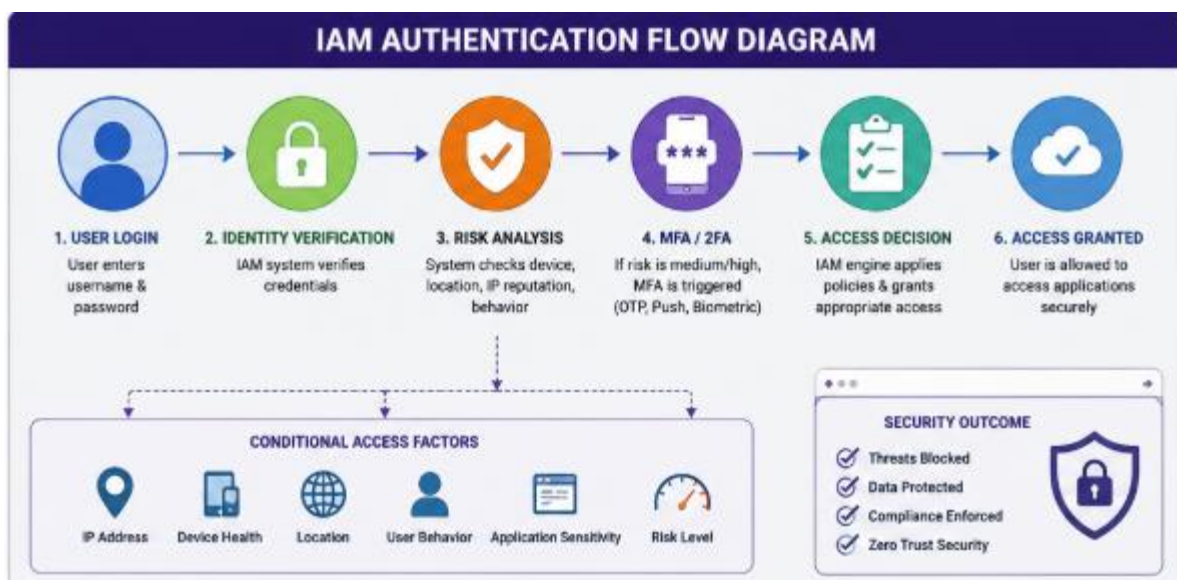
Geo-location

Risk Analysis

MFA

Conditional Access

Output/Screenshot



Result

Successfully analyzed IP-based IAM authentication workflow.

Learning Outcome

Learned how modern IAM systems secure user access.

Important Notes for IAM Analysts

Used in:

- Microsoft Entra ID
- Okta
- AWS IAM
- Zero Trust architectures

TASK 12 — FIREWALL & NETWORK SECURITY ANALYSIS

Task Name

Enterprise Firewall Network Security Analysis

Objective

To understand how firewalls secure enterprise networks.

Steps Performed

Step 1

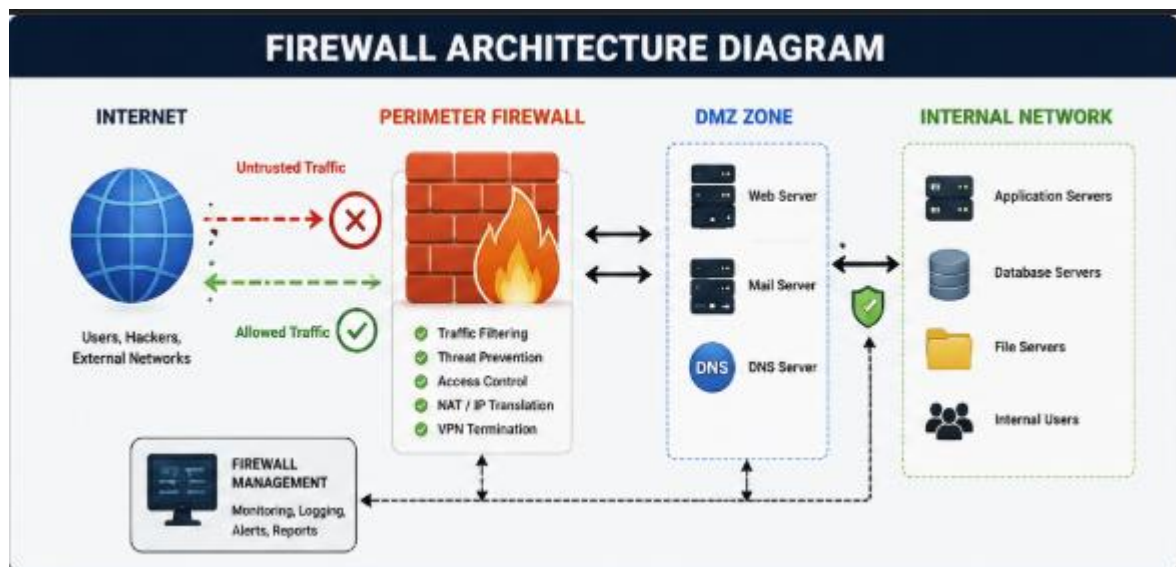
Created firewall architecture diagram.

Step 2

Mapped:

Internet → Firewall → VPN → Internal Network

Output/Screenshot



Result

Successfully analyzed firewall-based network protection.

Learning Outcome

Learned how organizations secure internal infrastructure.

Important Notes for IAM Analysts

Firewalls help IAM/security teams:

- Restrict unauthorized traffic
- Protect authentication systems
- Monitor suspicious access